

Crop Recommendation System Using Machine Learning Using k-nearest neighbors algorithm

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ABSTRACT

Agriculture remains a crucial sector, supporting nearly 58% of the global population. However, the majority of farmers still rely on traditional practices and experience rather than data-driven approaches, leading to inefficient crop selection and financial uncertainty. To bridge this gap, we present a machine learning-based crop recommendation system that considers weather conditions and soil type to suggest the most suitable crops. Our model incorporates multiple machine learning algorithms, including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes, Decision Tree, Logistic Regression, XGBoost, AdaBoost, and Gradient Boosting. Key parameters such as temperature, humidity, rainfall, nitrogen, soil pH, and nutrient content are considered due to their direct influence on crop yield. To ensure ease of use, we have developed a Python-based web interface that allows farmers to input soil and environmental parameters. The system then analyzes these inputs and provides personalized crop recommendations based on the trained models. The dataset used in this research has been sourced from Kaggle and the Food and Agriculture Organization of India, ensuring a robust and region-specific prediction model. This approach aims to promote precision agriculture, enhance productivity, and reduce financial risks for farmers by providing data-driven decision support.

Keywords—Machine Learning, Crop Recommendation, KNN, Decision Tree, Naive Bayes, Random Forest, Gradient Boosting, AdaBoost, XGBoost, Crop Yield Prediction.

1. INTRODUCTION

Agriculture is one of the most critical sectors in the global economy, providing food security and livelihood to millions. However, traditional farming practices often rely on intuition and past experiences, which agricultural decision-making through data-driven insights.

may not always lead to optimal crop selection and yield. With rapid advancements in technology, particularly in machine learning (ML) and artificial intelligence (AI), there is an opportunity to transform

This study proposes a Crop Recommendation System (CRS) that utilizes machine learning techniques to suggest the most suitable crop for specific environmental conditions. The system processes crucial agricultural factors such as soil nutrients (Nitrogen, Phosphorus, and Potassium), temperature, humidity, pH levels, and rainfall, which significantly influence crop selection. The proposed model incorporates various machine learning algorithms, including K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), XGBoost, AdaBoost, and Gradient Boosting. The dataset used for model training is obtained from authentic sources such as Kaggle and the Food and Agriculture Organization of India, ensuring accurate and location-specific recommendations. The system is implemented using Python and Flask, providing a user-friendly interface where farmers can input their environmental conditions and receive suitable crop recommendations.

This research aims to bridge the gap between traditional farming practices and modern AI-driven agriculture, ensuring precision farming and improved productivity. By leveraging machine learning, the system helps farmers optimize crop selection, reduce resource wastage, and increase overall agricultural efficiency. The following sections discuss related work, methodology, experimental results, and future research directions.

2.RELATED WORKS

[1]Farida Siddiqi Prity, MD.Mehadi Hasan, Shakhawat Hossain Saifl, Md.Maruf Hossain, Sazzad Hossain Bhuiyanl, Md. Ariful Islam, Md Tousif Hasan Lavlu “Enhancing Agricultural Productivity: A Machine Learning Approach to Crop Recommendations”. explores how machine learning (ML) can optimize crop selection to improve agricultural productivity. It addresses challenges like climate variability, soil conditions, and traditional farming limitations. The study evaluates nine ML models, including Logistic Regression, SVM, Decision Tree, and Random Forest, using data set on climate, soil nutrients. Random Forest achieved the highest accuracy (99.31%). The system provides data-driven crop recommendations, helping farmers make informed decisions, increase prediction and ensure sustainable farming. This approach supports food security and economic growth in the agricultural sector.

[2] Shafiulla Shariff, Shwetha R B, Ramya O G, Pushpa H, Pooja K R “Crop Recommendation using Machine Learning Techniques” project aims to assist farmers in selecting the most suitable crop based on soil characteristics. It employs KNN, Decision Tree, Naïve Bayes, Random Forest, and Gradient Boosting algorithms to analyze soil parameters such as pH, humidity, and NPK levels. The system enhances agricultural decision-making by providing data-driven crop recommendations, ensuring better productivity and sustainable farming practices.

[3]Dhruvi Gosai, Chintal Raval, Rikin Nayak, Hardik Jayswal, Axat Patel “Crop Recommendation System using Machine Learning” automates crop selection, implemented to train and test crop classification models. The performance of

reducing farmers' reliance on traditional knowledge. It utilizes IoT sensors for real-time soil analysis and applies ML algorithms like Random Forest and XGBoost for accurate recommendations. Additionally, CNN-based plant disease detection enhances monitoring, but dependence on technology may limit manual assessment skills.

[4] Ms. Sarika Gambhir, Manish Sharma, Khushboo Agarwal, Keshav Kumar, Lakshya Kumar, Mayank Chaudhary “Crop Recommendation System Using Machine Learning” helps farmers choose suitable crops based on environmental and economic aspects. It uses datasets to analyze rainfall, temperature, area, and soil quality. It applies machine learning models to predict yield, recommend crops, and guide fertilizer timing. This supports productivity, reduces losses, and improves efficiency.

3.METHODOLOGY

This paper outlines the structured approach taken to develop the crop recommendation system using machine learning and deep learning techniques. The system is designed to analyze environmental and soil parameters to predict the most suitable crops for a given location.

The process begins with data collection and preprocessing, where a dataset containing key agricultural factors such as nitrogen, phosphorus, potassium, temperature, humidity, pH levels, and rainfall is cleaned and standardized. This ensures that the data is formatted correctly for training machine learning models.

Next, multiple supervised learning algorithms such as k-nearest neighbors,

these models is evaluated using various metrics, including accuracy, precision, recall, and F1-score.

Finally, the trained model is deployed using Flask, allowing users to input their soil parameters via a web interface and receive real-time crop recommendations with crop yield predictions. This methodology ensures scientific accuracy, efficiency, and practical applicability in modern agriculture.

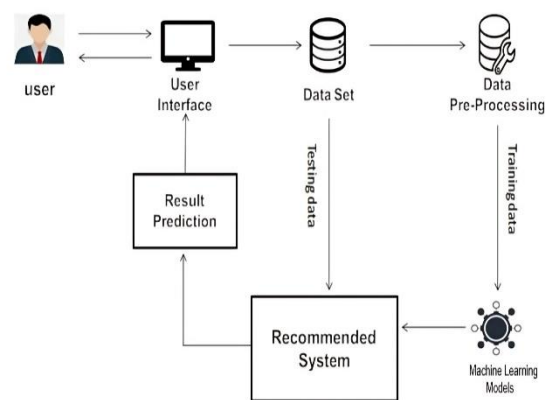


Fig.1 Architecture

3.1 DATA COLLECTION & PREPROCESSING

3.1.1 Data Collection

The dataset used in this research is sourced from Kaggle and the Food and Agriculture Organization of India, containing detailed agricultural data. The dataset includes essential features such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH levels, and rainfall, which are critical factors for crop selection. The dataset is stored in CSV format (crop_recommendation.csv) and is read using pandas in the train_knn.py script. The features are extracted as numerical inputs, and the crop labels are stored as categorical data.

deployment in the Flask-based web application.

3.1.2 Data Preprocessing

To ensure data quality and consistency, various preprocessing techniques are applied:

I Handling Missing Values

The dataset is checked for missing or null values, and imputation techniques are applied if necessary.

II Encoding Crop Labels

Since the crop names are categorical, they are converted into numerical labels using Ordinal Encoding. This transformation is necessary for training machine learning models.

III Feature Scaling

Since different features have varying scales (e.g., nitrogen values range from 0 to 200, while pH values range from 0 to 14), StandardScaler is applied to normalize the data. Standardizing helps models such as logistic regression, SVM, and deep learning models converge faster.

IV Train-Test Splitting

The dataset is divided into 80% training and 20% testing to evaluate model performance. The train-test split ensures that models generalize well on unseen data.

3.1.3 Data Integration with Machine Learning Models

After preprocessing, the preprocessed dataset is used to train different machine learning and deep learning models, including K-Nearest Neighbors (KNN), Decision Tree, Random Forest, SVM, Naïve Bayes, Logistic Regression, XGBoost, and AdaBoost. Additionally, deep learning architectures such as LSTM and GRU are explored for improved accuracy. The trained models are stored using joblib for it more

3.1.4 Real-Time Input Handling via Flask

In flask application, user inputs for N, P, K, temperature, humidity, pH, and rainfall are received through an HTML form. The inputs are scaled before making predictions using the trained model.

3.2 MACHINE LEARNING MODELS

Machine learning algorithms play a crucial role in predicting suitable crops based on environmental and soil conditions. This research explores multiple machine learning models to determine the best-performing model for crop recommendation. The selection of these models is based on their ability to handle classification tasks effectively while ensuring optimal accuracy and efficiency.

3.2.1 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a simple yet effective classification algorithm that assigns a class label to a given sample based on the majority class among its 'k' nearest neighbors. The performance of KNN heavily depends on the value of 'k' and the chosen distance metric.

$$d(p, q) = \sqrt{\sum (p_i - q_i)(p_i - q_i)}$$

3.2.2 Decision Tree Classifier

Decision Trees are hierarchical models that split data into subsets based on feature values, forming a tree structure where each node represents a feature, and leaves represent class labels. They are effective in handling non-linearity in datasets.

3.2.3 Random Forest Classifier

Random Forest is an ensemble learning method that builds multiple decision trees and merges their outputs to improve accuracy and prevent overfitting. The model selects random subsets of data and features, making effective in handling complex datasets with high dimensionality.

robust compared to individual decision trees.

3.2.4 Support Vector Machine (SVM)

SVM is a powerful classification algorithm that identifies the optimal hyperplane that separates different classes in a high-dimensional space. The choice of kernel function (linear, polynomial, or radial basis function) plays a significant role in determining the model's performance.

3.2.5 Logistic Regression

Logistic Regression is a statistical model that estimates the probability of a categorical outcome based on input variables. It is commonly used for binary classification but can be extended to multi-class problems using the softmax function.

3.2.6 Naive Bayes Classifier

Naive Bayes is a probabilistic classifier based on Bayes' theorem, assuming independence among features. Despite its simplicity, it performs well for text classification and categorical datasets.

3.2.7 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is an optimized boosting algorithm that sequentially improves weak learners. It is widely used in Kaggle competitions due to its superior performance.

3.2.8 AdaBoost Classifier

AdaBoost (Adaptive Boosting) combines multiple weak classifiers iteratively to enhance prediction accuracy. It assigns higher weights to misclassified instances to improve subsequent iterations.

3.2.9 Gradient Boosting Classifier

Gradient Boosting improves model performance by sequentially minimizing residual errors of previous models. It is

3.4 MODEL EVALUATION METRICS

To ensure the reliability and effectiveness of the crop recommendation system, multiple evaluation metrics are utilized to assess the performance of the implemented models. These metrics provide a comprehensive understanding of the model's classification ability and its capacity to generalize across different crop categories.

3.4.1 Accuracy Score

The accuracy score is one of the fundamental evaluation metrics, representing the proportion of correct predictions out of the total number of predictions. It is calculated as follows:

Where:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{TN} + \text{FN})$$

1. **TP (True Positives):** Correctly predicted instances of a particular crop.
2. **TN (True Negatives):** Correctly predicted non-relevant crops.
3. **FP (False Positives):** Incorrectly predicted crop recommendations.

FN (False Negatives): Relevant crop instances that were not correctly classified.

3.4.2 Precision, Recall, and F1-Score

To further evaluate the effectiveness of the models, precision, recall, and F1-score are analyzed:

- **Precision** measures how many of the predicted crop recommendations were actually correct:

A high precision score indicates fewer false positives, meaning that the model does not incorrectly recommend crops frequently.

- **Recall (Sensitivity)** evaluates the model's ability to correctly classify all relevant crop recommendations:

A higher recall value ensures that most relevant crops are correctly identified, reducing false negatives.

- **F1-Score** is the harmonic mean of precision and recall, balancing both metrics:

A high F1-score ensures that the model performs well in both precision and recall, making it an ideal metric for imbalanced datasets.

3.4.3 Confusion Matrix

The confusion matrix is an essential tool that visualizes the model's classification performance by showing actual versus predicted values. It helps in understanding misclassification patterns:

Actual/Predicted (Crop A)	Positive Negative(Other Crops)	
	TP	FN
Positive (Crop A)	TP	FN
Negative (Other Crops)	FP	TN

By analyzing the confusion matrix, one can determine whether the model is biased towards a specific crop or whether certain crops are more prone to misclassification.

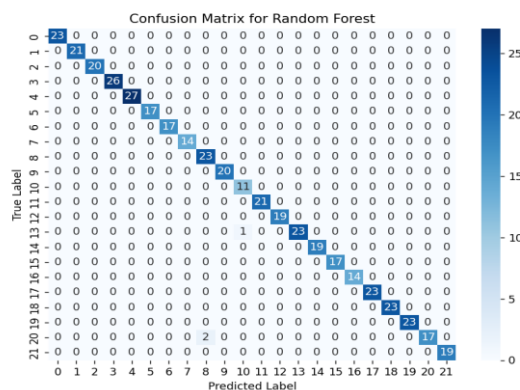


Fig.2 Random Forest

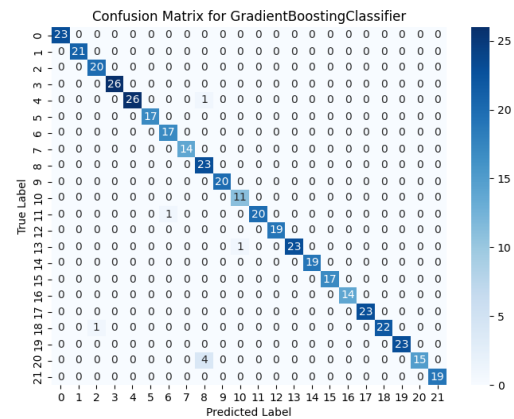


Fig.3.GradientBoost

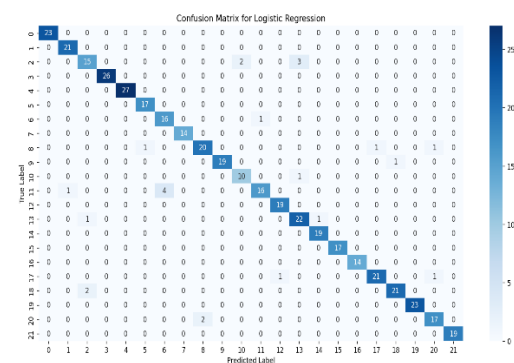


Fig.4 LogisticRegression

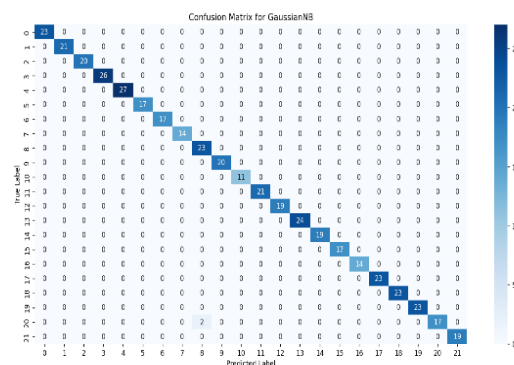


Fig.5 GussianNB

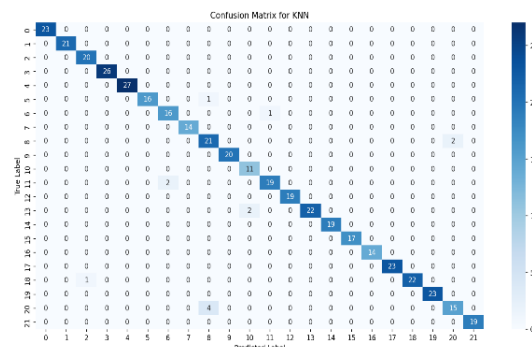


Fig.6 KNN

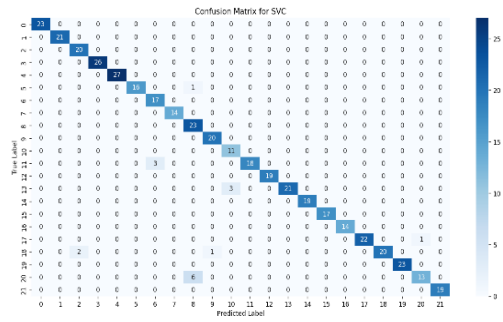


Fig.7 SVC

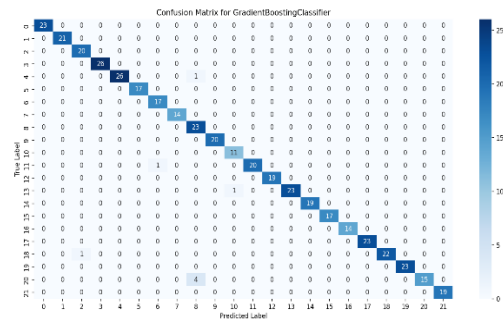


Fig.8 GradientBoost

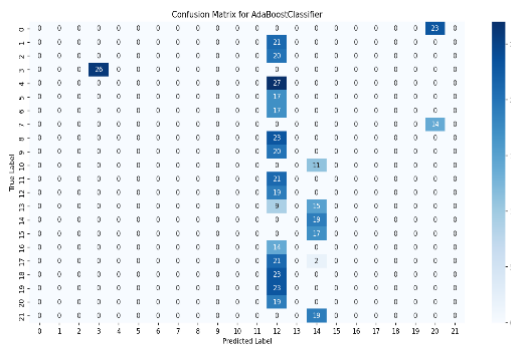


Fig.9 AdaBoost

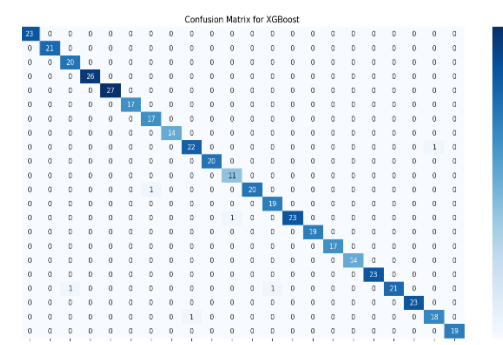


Fig.10 XGBoost

4. RESULTS & DISCUSSION

4.1 CONFUSION MATRIX COMPARISON

The confusion matrix allows a direct comparison of machine learning model performance. Random Forest and XGBoost exhibit lower misclassification rates than traditional models like Decision Trees and KNN. Boosting models (XGBoost, AdaBoost, and Gradient Boosting) significantly reduce false positives and false negatives, demonstrating higher predictive stability. These findings confirm that ensemble learning models outperform traditional ML models in multi-class classification tasks.

4.2 Precision-Recall Curve

The Precision-Recall Curve for multi-class classification shows how well each crop class is predicted across different recall values. Most classes maintain high precision near 1.0 for a majority of recall values, meaning the model confidently predicts correct classifications. Some fluctuations appear at high recall values for certain classes, indicating possible misclassification or overlapping crop feature spaces. The spread of lines suggests variability in precision among different crops, meaning some crops are easier to classify correctly than others. Precision drops sharply at the highest recall values for a few crops, which could indicate difficulty in distinguishing between similar crops under certain conditions.

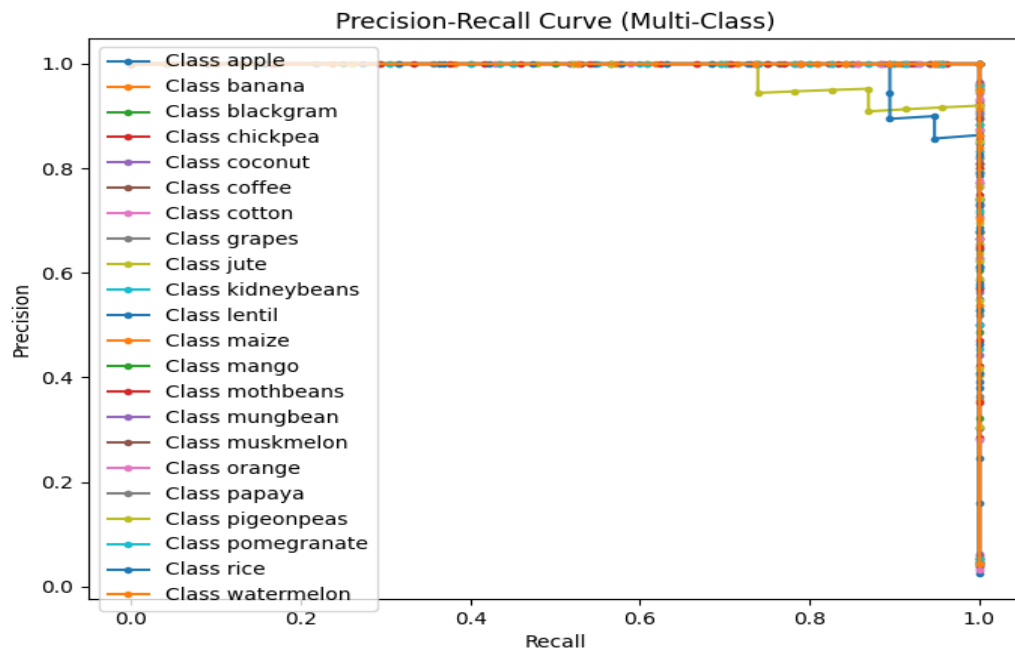


Fig.11 precision-recall curve

This image represents a Precision-Recall (PR) Curve for a Multi-Class Classification problem in the crop recommendation system. The PR curve is used to evaluate the performance of a classification model, particularly in cases where the dataset is imbalanced.

Description of the Graph:

1. **X-Axis (Recall):**
 - Measures the proportion of actual positive cases correctly identified by the model.
 - Higher recall means the model is correctly identifying most of the relevant classes.
2. **Y-Axis (Precision):**
 - Measures how many of the predicted positive cases are actually correct.
 - Higher precision means fewer false positives.
3. **Multiple Colored Lines:**
 - Each line represents a different crop class, such as apple, banana, rice, maize, etc.
 - The precision-recall trade-off for each crop classification is shown.
4. **Key Observations:**
 - Most classes have high precision and recall values, indicating good model performance.
 - Some minor fluctuations suggest variations in classification confidence across different crop classes.

4.3 ROC Curve

The Receiver Operating Characteristic (ROC) Curve evaluates how well the model distinguishes between different crop classes. All crop classes have an AUC (Area Under Curve) of 1.00, indicating perfect classification for each class. The ROC curves stay close to the top-left corner, suggesting that the model rarely produces false positives. The true positive rate (sensitivity) is maximized at all points, meaning the model detects each crop class with near-perfect accuracy. The lack of deviation from the diagonal baseline confirms that the model does not struggle in distinguishing between crop categories. This suggests exceptional classification performance, potentially due to an overfitted training set or a well-optimized learning approach.

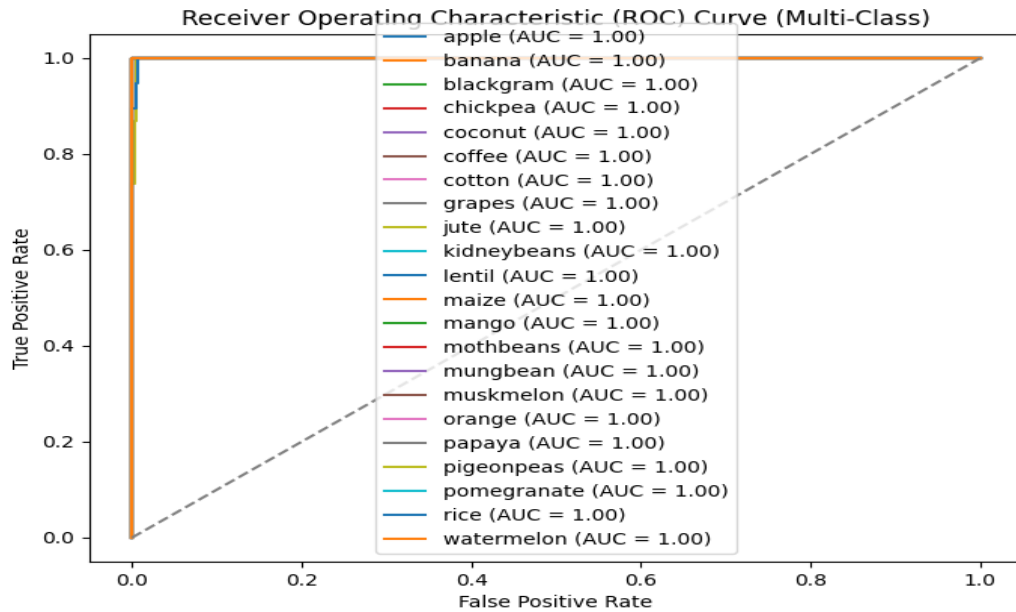


Fig.12 ROC-curve

This image represents a Receiver Operating Characteristic (ROC) Curve for a Multi-Class Classification model used in the crop recommendation system. The ROC curve is a standard tool for evaluating classification performance by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR).

Breaking Down the Graph:

- X-Axis (False Positive Rate - FPR):**
 - Measures the proportion of negative samples incorrectly classified as positive.
 - Lower FPR is better because it indicates fewer false alarms.
- Y-Axis (True Positive Rate - TPR):**
 - Measures how many actual positive samples were correctly identified.
 - Higher TPR is better as it indicates better classification.
- Multiple Colored Lines (Crop Classes):**
 - Each line represents the ROC curve for a different crop class (apple, banana, rice, maize, etc.).
 - All lines are reaching the upper-left corner (TPR = 1, FPR = 0), showing high performance.
- Area Under the Curve (AUC = 1.00 for All Classes):**
 - The AUC score quantifies the classifier's ability to distinguish between classes.
 - AUC = 1.00 means the model achieves perfect classification for every crop type.

4.4 Model Performance Comparison

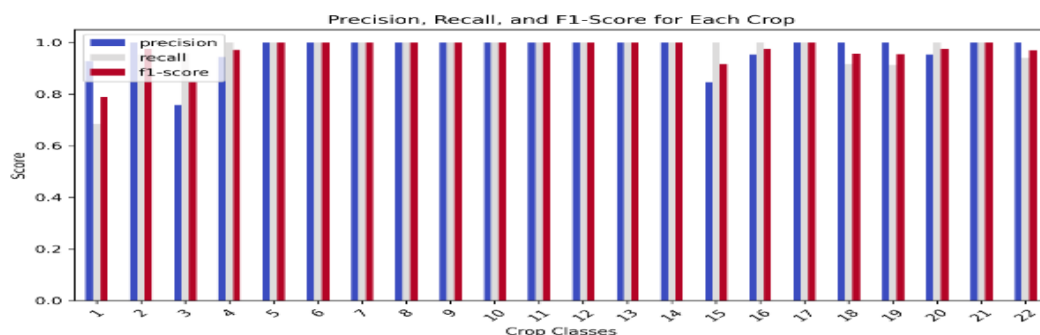


Fig.13 Comparison of precision, recall and f1-score for each crop

This bar chart visualizes the precision, recall, and F1-score for each crop class, indicating the model's classification performance. High scores suggest effective classification, while variations highlight challenges in certain crop predictions.

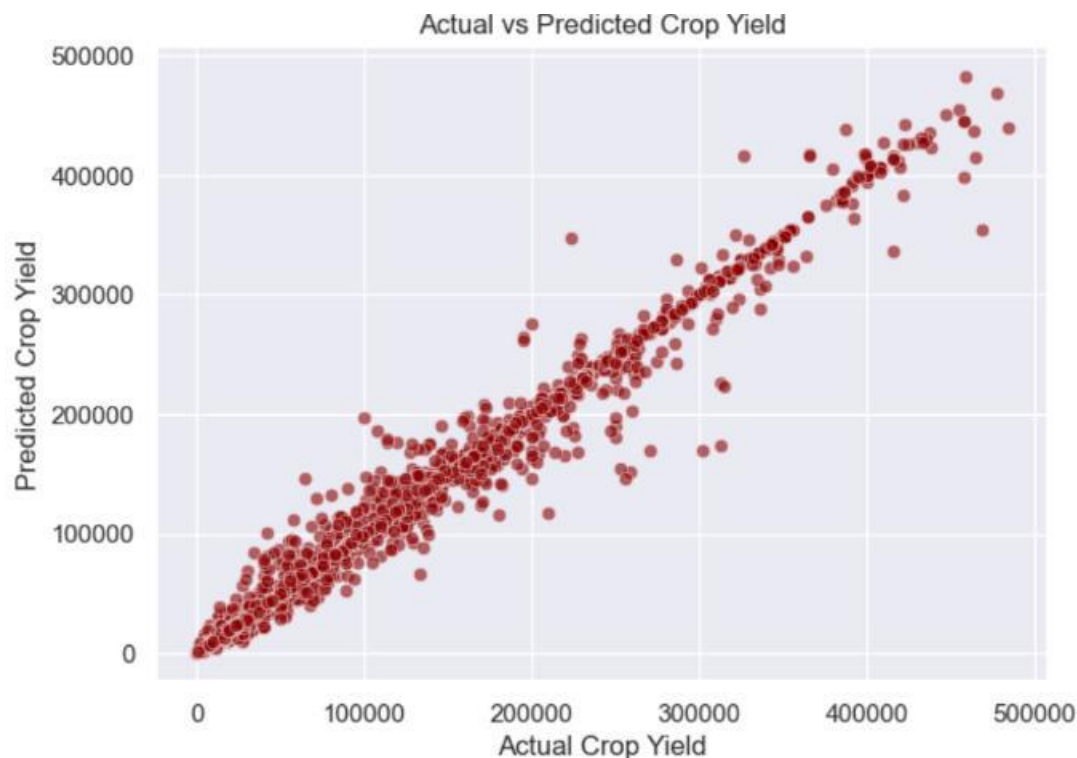


Fig.14 Actual vs Predicted Crop Yield

The scatter plot represents the relationship between actual and predicted crop yield values. The x-axis denotes the actual crop yield, while the y-axis represents the predicted crop yield generated by the model. Each red dot signifies a data point, showing how closely the predictions align with the actual values. The strong clustering of points along a diagonal trend indicates a high correlation between actual and predicted values, suggesting that the model performs well in forecasting crop yield. However, some outliers are visible, where predictions deviate from actual values, highlighting areas where the model may need further refinement.

The accuracy results of different machine learning models used in our Crop Recommendation System indicate their effectiveness in predicting suitable crops based on soil and climate conditions. Among the models, Naïve Bayes (99.54%), Random Forest (99.31%), Decision Tree (98.63%), and XGBoost (98.60%) demonstrated exceptional accuracy, making them the most reliable choices for this task. These models efficiently analyze features such as nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall to provide optimal crop recommendations. Gradient Boosting (98.18%) and K-Nearest Neighbors (97.04%) also performed well, showcasing their ability to classify crops accurately. Support Vector Machine (96.13%) and Logistic Regression (94.54%) exhibited decent accuracy, though they may struggle with capturing complex patterns in the data. However, AdaBoost (14.54%) significantly underperformed, possibly due to its sensitivity to data variations, making it less suitable for this application. The overall findings highlight the effectiveness of ensemble learning models such as Random Forest and XGBoost in delivering highly accurate

recommendations. This study confirms that machine learning can significantly enhance precision agriculture, allowing farmers to make data-driven decisions for better yield and resource management. Further refinements, such as real-world validation and integration with dynamic environmental data, could further improve the system's practical applicability.

Model	Accuracy(%)
KNN	97.04
Decision Tree	98.63
Random Forest	99.31
Naive Bayes	99.54
Logistic Regression	94.54
SVM	96.13
AdaBoost	14.54
Gradient Boosting	98.18
XGBoost	98.6

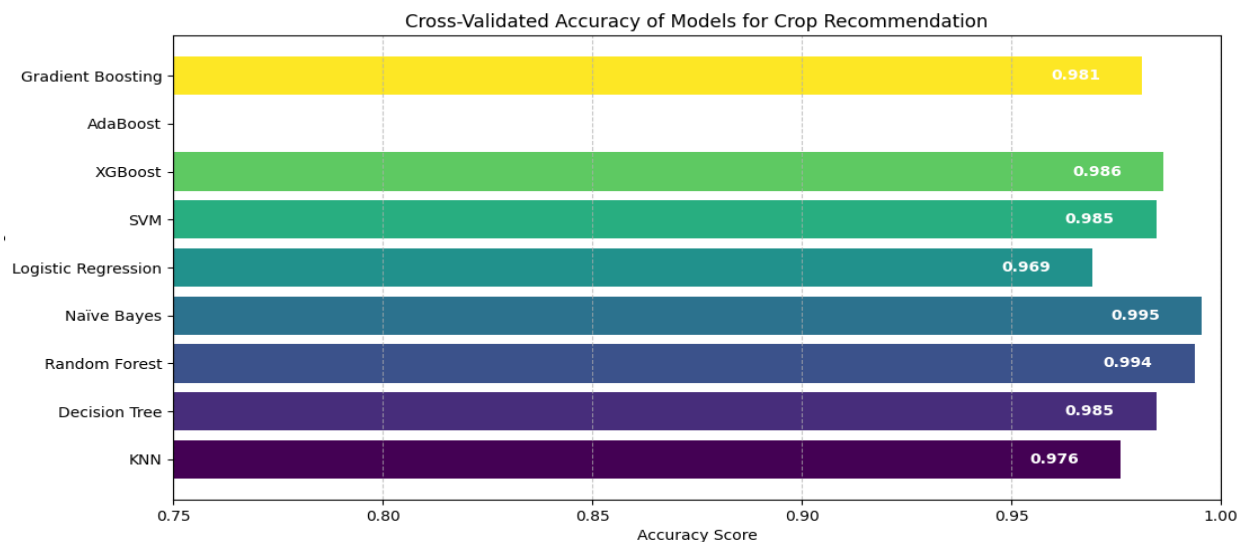


Fig.15 Accuracy of Models for Crop Recommendation

5.CONCLUSION

Our crop recommendation system utilizes the K-Nearest Neighbors (KNN) algorithm, achieving an impressive accuracy of **97.04%**. KNN effectively analyzes soil nutrients and environmental factors to identify the most suitable crop, making it a reliable and practical solution for precision agriculture.

Unlike complex ensemble models that require extensive fine-tuning, KNN strikes a balance between simplicity and accuracy. It classifies crops based on similarities in past data, ensuring precise recommendations without high computational costs. This makes it ideal for real-world farming applications, where quick and efficient decision-making is essential.

A key advantage of KNN is its adaptability. As a non-parametric, instance-based algorithm, it updates dynamically, allowing real-time adjustments as conditions change. This ensures that

farmers receive the most relevant and timely recommendations, enhancing productivity and resource management.

By integrating real-time data and optimizing K-values, KNN can further improve performance. Our project delivers a farmer-friendly, efficient, and accessible approach to smart farming, maximizing crop yield and sustainability.

6. FUTURE SCOPE

Integration with IoT Devices: Incorporating real-time soil and climate data for dynamic crop recommendations. Mobile Application Development: Extending the system as a user-friendly mobile application for easy access by farmers. Expanding Geographical Coverage: Adapting the model for diverse soil types and climate conditions across different regions. Improved Model Interpretability: Enhancing explainability of predictions using SHAP (SHapley Additive exPlanations) values. By leveraging machine learning and AI-driven decision-making, this research paves the way for precision agriculture, enabling efficient land use, higher crop yields, and sustainable farming practices. The implementation of such intelligent systems revolutionize the agricultural sector, ensuring better productivity and economic benefits for farmers worldwide.

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